

STRATHMORE AGDM, AB, Canada

WMO: 715260

Lat: 51.0387N

Lon: 113.2897W

Elev: 3202

StdP: 13.07

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.1	-12.7	-23.7	1.7	-17.8	-18.0	2.3	-12.4	25.7	27.3	23.0	23.2	7.4	210	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	83.4	61.7	80.1	60.4	77.0	59.3	65.1	77.4	63.0	75.4	61.2	73.4	8.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	89.0	70.8	58.2	81.6	68.6	56.2	75.8	66.5	31.8	77.5	30.1	75.5	28.8	73.6	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	21.0	18.6	DB	-24.7	90.1	4.1	3.2	-27.6	92.4	-30.0	94.2	-32.3	96.0	-35.3	98.3	(4)
(5)			WB	-25.0	69.3	4.0	2.5	-27.9	71.1	-30.2	72.5	-32.5	73.9	-35.4	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	19.3	17.0	27.7	39.5	51.1	57.7	63.4	61.0	53.1	41.1	27.3	17.4		
	DBStd	19.88	14.81	14.51	13.57	9.26	7.78	5.67	5.11	5.85	8.44	9.00	12.87	14.41		
	HDD50	5036	953	925	691	328	81	6	0	3	62	294	682	1011		
	HDD65	9296	1418	1345	1156	765	433	228	92	148	364	739	1132	1476		
	CDD50	1302	0	0	1	13	115	238	415	344	155	20	1	0		
	CDD65	85	0	0	0	0	2	9	42	25	7	0	0	0		
	CDH74	1375	0	0	0	6	80	147	542	425	171	4	0	0		
	CDH80	290	0	0	0	0	6	23	127	98	36	0	0	0		
	WSAvg	8.9	9.1	8.7	9.3	10.0	9.8	9.1	8.2	7.8	8.6	9.0	9.1	8.6		
	PrecAvg	20.3	0.7	0.6	1.0	1.5	2.7	3.9	2.9	2.5	1.8	1.0	0.9	0.8		
(15) Precipitation	PrecMax	26.7	1.5	1.3	2.1	3.4	4.3	10.0	6.3	4.5	4.3	2.0	2.4	1.7		
	PrecMin	12.9	0.2	0.1	0.3	0.6	0.9	1.7	1.1	0.5	0.3	0.3	0.2	0.2		
	PrecStd	2.7	0.3	0.3	0.4	0.7	1.0	1.5	1.2	0.9	0.9	0.4	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.9	49.1	60.0	72.0	80.5	83.3	87.8	87.2	84.3	71.5	57.9	50.4		
		MCWB	38.3	38.9	44.7	50.1	55.0	62.1	64.0	60.8	58.2	52.1	44.1	37.7		
	2%	DB	43.8	43.2	54.4	66.0	76.1	78.0	83.7	82.5	79.0	65.1	51.7	43.8		
		MCWB	36.3	35.7	41.4	47.4	53.8	58.6	63.6	61.0	56.6	49.3	40.9	36.6		
	5%	DB	41.0	39.6	49.5	60.3	71.6	74.3	79.8	79.0	74.2	60.1	47.2	40.2		
		MCWB	34.8	33.5	38.8	44.5	51.9	57.5	63.1	60.2	55.2	46.7	38.1	34.0		
	10%	DB	38.4	35.9	44.6	55.2	67.2	70.8	76.5	74.9	68.9	55.4	43.0	36.5		
		MCWB	33.3	31.2	36.6	41.9	50.0	55.3	61.5	59.2	53.2	44.7	35.6	31.4		
	0.4%	WB	39.8	39.4	45.9	51.4	57.1	64.4	69.8	65.7	60.4	53.8	45.3	40.0		
		MCDB	46.7	46.8	57.6	70.2	74.7	76.6	80.6	78.2	79.0	68.9	57.3	46.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.0	36.2	42.3	48.1	54.9	61.3	66.5	63.5	57.9	50.5	41.7	36.9		
		MCDB	42.7	42.7	52.7	63.8	72.9	73.9	78.3	76.9	74.1	62.6	50.4	43.4		
	5%	WB	35.3	33.9	39.5	45.5	53.0	59.0	64.4	61.7	55.9	47.7	39.0	34.2		
		MCDB	40.6	39.1	48.5	58.8	69.0	70.9	76.4	74.8	71.5	58.6	46.4	39.9		
	10%	WB	33.1	31.3	36.8	42.7	51.0	56.9	62.5	59.9	54.0	45.2	36.1	31.5		
		MCDB	38.3	35.4	44.0	53.4	64.6	67.8	74.4	72.6	67.9	54.3	42.2	36.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.9	16.7	17.7	20.6	22.2	20.1	22.4	22.6	22.3	20.2	17.3	16.0		
		MCDBR	18.7	20.3	25.1	30.6	29.4	26.4	28.0	30.3	31.5	28.3	23.0	18.6		
		MCWBR	12.9	14.4	14.5	15.2	12.0	11.5	12.6	12.7	13.0	14.8	14.3	13.4		
		MCDBR	17.8	19.6	24.0	28.9	26.9	23.0	25.1	26.7	29.0	26.9	22.6	18.8		
	5% WB	MCWBR	12.6	14.2	14.8	14.8	11.4	11.4	12.4	12.2	12.5	14.4	14.8	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.264	0.280	0.330	0.347	0.342	0.342	0.358	0.315	0.278	0.251	0.235		
	taud		2.393	2.449	2.478	2.348	2.355	2.410	2.423	2.372	2.459	2.541	2.494	2.410		
	Ebn at Noon		252	282	297	289	288	290	287	276	279	273	255	245		
	Edh at Noon		19	23	28	35	37	35	34	34	28	21	17	16		
(44) All-Sky Solar Radiation	RadAvg		366	685	1113	1517	1825	1974	2127	1711	1208	736	397	281		
	RadStd		28	61	89	113	124	121	105	116	119	85	34	26		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (78 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page